Muhammad Ahsan Aftab

Full-Stack Engineer

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Summary

Full-stack developer with a BS in Computer Science (Air University Islamabad, 2026). Builds web and mobile applications end to end: React and Flutter frontends, Node.js, Django, and FastAPI backends on PostgreSQL, MongoDB, and Supabase, with real-time features over WebSockets and LLM-powered features where they fit. 1st Position at NDMA Pakistan's Disaster Early Warning Tech Expo 2025. Seeking a full-stack engineering role to contribute to production applications while continuing to grow.

Technical Skills

Languages: TypeScript, JavaScript, Python, Dart, SQL, Solidity, C++
Frontend & Mobile: React.js, Next.js, Material UI, Tailwind CSS, Flutter, HTML/CSS, Responsive UI Design, State Management
Backend: Node.js, Express.js, Django REST Framework, FastAPI, Flask, REST APIs, WebSockets
Databases & BaaS: PostgreSQL, MySQL, MongoDB, Supabase, Firebase, Cloud Firestore
DevOps & Tools: Git, Docker, Render, Vercel, Postman, n8n, Hugging Face Spaces
AI Integration: LLMs, Gemini, Vertex AI, Google ADK, Multi-Agent Systems, Prompt Engineering, PyTorch, OpenCV

Experience

Full Stack Web Development Intern **Jul 2025 – Aug 2025**
Octathorn *Rawalpindi, Pakistan*
- Built real-time communication features using WebSockets, enabling live messaging and synchronized multi-user interaction.
- Developed reusable, responsive React.js and Material UI components and wired frontend workflows to backend REST APIs.

App Development Intern **Jul 2025 – Sep 2025**
AKSA-SDS *Hybrid – Islamabad, Pakistan*
- Developed cross-platform Flutter applications with Firebase integration, REST API connectivity, and responsive UI flows.
- Implemented state management, reusable screens, and a scalable app structure within a supervised hybrid team.

Web and Mobile App Developer Intern **Jul 2024 – Sep 2024**
Laser & Electro Optical Solutions (LEOS) *Pakistan*
- Built and maintained a full-stack Flutter app for web and mobile, using Firebase and Cloud Firestore.

Projects

Weather-Aware Truck Routing | *Django REST Framework, React, TypeScript, MapLibre* *Solo project*
- Designed a Django REST API that finds up to three truck routes, estimates arrival time at each checkpoint, and scores the hourly forecast for that place and time from Low to No Travel, taking load weight into account.
- Built the React and TypeScript map UI (MapLibre GL JS, TanStack Query) with route comparison, a weather heatmap, and a time slider; thinned 15,000-point routes to about 1,000 points for the browser.

CIRO – Crisis Intelligence & Response Orchestrator | *FastAPI, Supabase, Google ADK, Flutter, React* *Team project*
- Built the FastAPI backend, which orchestrates a five-agent Gemini pipeline (collect, detect, reason, plan, simulate) that turns citizen reports into prioritized response plans.
- Stored each report's detected language, confidence score, and agent trace in Supabase; containerized the backend with Docker and contributed to the Flutter and React clients.

Miras – Digital Archiving & Marketplace | *Solidity, Express.js, React, MongoDB, Docker* *Team project*
- Wrote the Solidity smart contracts for NFT minting, on-chain provenance tracking, and marketplace listings and transactions, connected to the Node.js backend with Ethers.js.
- Added phase-based email updates for listing submissions (Resend) and domain whitelisting to the Express backend; containerized the React client with Docker; live at miraas.tech.

PDME – Pakistan Disaster Management Ecosystem | *Node.js, PostgreSQL, Python* *1st Place, NDMA DEWTE 2025*
- Built a Node.js backend connected to PostgreSQL for flood and wildfire risk analysis and emergency coordination.
- Trained the flood and forest-fire prediction models in Python.

AI-Driven Digital Artifact Restoration | *PyTorch, SegFormer, GANs, OpenCV* *Solo project*
- Wrote an OpenCV damage generator for training data, fine-tuned SegFormer-B2 for 6-class damage segmentation, and trained an inpainting GAN evaluated with PSNR; deployed both models as APIs on Hugging Face Spaces.

Education

Air University Islamabad **2022 – 2026**
BS Computer Science *CGPA: 3.17*

Achievements & Certifications

- **1st Position, NDMA Disaster Early Warning Tech Expo (DEWTE) 2025** – National Disaster Management Authority, for the PDME project.
- **IBM JavaScript Programming Essentials** – IBM SkillsBuild | **Version Control** – Meta